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Mechanical Thrombectomy in Ischemic Stroke With a Medium or Distal Arterial Occlusion

The DISCOUNT Randomized Clinical Trial

Frédéric Clarençon, MD, PhD; Fouzi Bala, MD, PhD; Amandine Baptiste, MSc; Marco Pasi, MD, PhD; Gaultier Marnat, MD; Igor Sibon, MD, PhD; Benjamin Gory, MD, PhD; Sébastien Richard, MD, PhD; Raoul Pop, MD, PhD; Valérie Wolff, MD, PhD; Cédric Fauché, MD; Matthias Lamy, MD; Vincent Costalat, MD, PhD; Caroline Arquizan, MD; Christine Rodriguez-Régent, MD; Guillaume Turc, MD, PhD; Christophe Cognard, MD, PhD; Louis Fontaine, MD; Federico Di Maria, MD; Bertrand Lapergue, MD, PhD; Nicolas Bricout, MD; Hilde Henon, MD; Titien Tuilier, MD; Aymeric Wittwer, MD; Pierre Louis Alexandre, MD; Solène De Gaalon, MD; Laurent Spelle, MD, PhD; Christian Denier, MD, PhD; Emmanuel Chabert, MD; Anna Ferrier, MD; Aymeric Rouchaud, MD, PhD; Francisco Macian-Montoro, MD; Sébastien Soize, MD, PhD; Solène Moulin, MD, PhD; William Boisseau, MD; Candice Sabben, MD; Charlotte Barbier, MD; Marion Boulanger, MD, PhD; Omer Faruk Eker, MD, PhD; Tae-Hee Cho, MD, PhD; Jean-Christophe Gentric, MD, PhD; Serge Timsit, MD, PhD; Marc-Antoine Labeyrie, MD, PhD; Peggy Reiner, MD; Fernando Pico, MD, PhD; Isabelle Durand-Zaleski, MD, PhD; Sonia Alamowitch, MD; Agnès Dechartres, MD, PhD; for the DISCOUNT Investigators

IMPORTANCE Evidence regarding efficacy and safety of thrombectomy in acute ischemic stroke (AIS) due to medium or distal vessel occlusions (MDVOs) is lacking.

OBJECTIVE To evaluate the benefit of thrombectomy, in addition to medical treatment over medical treatment alone, in patients with an AIS related to a primary and isolated MDVO.

DESIGN, SETTING, AND PARTICIPANTS Randomized clinical trial conducted at 22 stroke centers in France from November 2021 to April 2025, with planned enrollment of 488 patients. The trial has been stopped after the planned interim analysis on the recommendation of the data and safety monitoring board for futility and increased rate of symptomatic intracranial hemorrhage with thrombectomy. Eligible adult patients had an AIS due to a primary MDVO within 8 hours of symptom onset or within 24 hours of last seen well if no hyperintense signal was present on fluid-attenuated inversion recovery imaging.

INTERVENTION Thrombectomy in addition to medical treatment (n = 123) or medical treatment alone (n = 121).

MAIN OUTCOMES AND MEASURES The primary end point was a good clinical outcome at 3 months, defined as a modified Rankin Scale score of 0 to 2, assessed by an independent, blinded assessor. Secondary end points included mortality rate at 3 months and adverse and serious adverse events.

RESULTS Of the 244 patients randomized (median age, 75 years [IQR, 67-81]; 56% male; median National Institutes of Health Stroke Scale score, 8 [IQR, 6-12]), 100 of the 123 patients in the thrombectomy group (81%) received thrombectomy and none of the 121 patients in the control group received thrombectomy; 217 (89%) completed follow-up. At 3 months, 72 of 116 patients (62%) in the thrombectomy group had a good clinical outcome vs 81 of 119 patients (68%) in the control group (odds ratio, 0.73 [95% CI, 0.40-1.31]; $P = .29$; adjusted absolute difference, -6.8% [95% CI, -19.4% to 5.7%]). The incidence of symptomatic intracranial hemorrhages was higher among the 100 patients who actually received thrombectomy than in those who did not (11% vs 3%, $P = .008$), as was incidence of subarachnoid hemorrhages (13% vs 2%, $P < .001$) and embolus migration (5% vs 1%, $P = .04$). Mortality rate did not significantly differ between the 2 groups (6% vs 8%; $P = .49$).

CONCLUSIONS AND RELEVANCE Thrombectomy did not lead to a higher rate of good clinical outcome at 3 months compared with medical treatment alone in patients with acute ischemic stroke related to an MDVO. Hemorrhagic complications were more frequent after thrombectomy.

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Author Affiliations: Author affiliations are listed at the end of this article.

Group Information: A list of the DISCOUNT Investigators appears in Supplement 6.

Corresponding Author: Frédéric Clarençon, MD, PhD, Department of Neuroradiology, Pitié-Salpêtrière Hospital, 47, Bd de l'Hôpital, 75013 Paris, France (frederic.clarencon@aphp.fr).

Mechanical thrombectomy has transformed the management of acute ischemic stroke (AIS), becoming the standard treatment^{1,2} for AIS related to large vessel occlusion (LVO). Randomized clinical trials (RCTs) have shown an absolute benefit in good clinical outcome ranging from 11% to 31% in favor of thrombectomy.³⁻⁹ However, these trials mainly enrolled patients with proximal occlusions, with only a small proportion of patients having more distal occlusions (0.5% to 14.3% involving M2 segments).¹⁰

Results from observational studies evaluating the benefit of thrombectomy in medium or distal vessel occlusions (MDVOs) have been equivocal. Some studies suggested a clinical benefit of thrombectomy,^{11,12} while others did not.¹³⁻¹⁵ Robust evidence supporting the utility of thrombectomy in primary MDVOs, which account for up to 20% of all AIS cases,¹⁶ is therefore lacking. On one hand, thrombectomy may facilitate recanalization and improve functional outcomes, as observed in proximal occlusions. On the other hand, thrombectomy in medium or distal vessels carries a higher risk of procedural complications such as perforation and intracranial hemorrhage, particularly subarachnoid hemorrhage, due to increased vessel tortuosity, fragility, and mobility compared with proximal ones.¹⁷ Furthermore, intravenous thrombolysis (IVT) alone achieves relatively high recanalization rates in MDVOs, reaching 40% to 50%.¹⁸ The benefit-risk balance of thrombectomy may, therefore, differ between medium or distal occlusions and LVOs.

Recently, 2 RCTs assessing the benefit of thrombectomy with best medical treatment compared with best medical treatment alone in MDVOs showed no significant difference in clinical outcomes.^{19,20} However, more recent unpublished RCTs (Oriental-MeVO²¹ and DISTALS²²), presented at the 2026 International Stroke Conference annual meeting, reported positive results for thrombectomy in MDVOs.

The aim of the DISCOUNT trial was to evaluate the efficacy of thrombectomy with best medical treatment, compared with best medical treatment alone, in achieving a good clinical outcome at 3 months in patients with AIS due to a primary MDVO.

Methods

Trial Design and Oversight

DISCOUNT was a multicenter open-label RCT with blinded assessment of the primary end point by an independent assessor. Recruitment was conducted in 22 French comprehensive stroke centers.

The trial protocol has been published²³ and was approved by the Comité de Protection des Personnes Ile de France 1 on July 8, 2021, and by the Agence Nationale de Sécurité du Médicament on July 12, 2021. All subsequent versions of the protocol were approved by the same ethics committee. The first and final versions of the protocol and protocol amendments are available in [Supplement 1](#). The statistical analysis plan is available in [Supplement 2](#). A list of the devices likely to be used in the trial at the discretion of the operator is available in [Supplement 3](#). All patients or their legal representatives pro-

Key Points

Question Does mechanical thrombectomy improve the rate of good clinical outcome in patients with acute ischemic stroke related to a primary and isolated medium or distal vessel occlusion?

Findings In this randomized clinical trial that included 244 participants, the rate of good clinical outcome (modified Rankin Scale score 0-2) at 3 months was 62% in the thrombectomy group and 68% in the medical treatment-alone group, a difference that was not statistically significant. The rate of symptomatic hemorrhage was more than 3 times higher with thrombectomy than with medical treatment alone (11% vs 3%).

Meaning In patients with acute ischemic stroke related to a medium or distal vessel occlusion, thrombectomy did not improve clinical outcomes and was associated with an increased risk of hemorrhage, which does not support its routine use in this setting.

vided written informed consent. If the patient was unable to consent at inclusion and his or her representative was not present, an emergency consent procedure was implemented, and a pursuit consent was signed as soon as the patient's condition allowed it or if not by his or her representative. This report fully adheres to the Consolidated Standards of Reporting Trials (CONSORT) reporting guidelines.

Participants

Patients were eligible if they were 18 years or older, had a symptomatic occlusion as evaluated by a National Institutes of Health Stroke Scale (NIHSS) score of 5 or greater or significant aphasia, were admitted within 8 hours from onset of symptoms or within 24 hours of last seen well if no hyperintense signal was present on fluid-attenuated inversion recovery acquisition and had an MDVO evaluated on computed tomography or magnetic resonance angiography. Based on neuroanatomical landmarks proposed by Fischer,²⁴ we defined a medium or distal vessel occlusion as an occlusion in 1 of the following segments: (1) M2 distal to the mid height of the insula or M3 segment of the middle cerebral artery (MCA); (2) A1, A2, or A3 segments of the anterior cerebral artery (ACA); and (3) P1, P2, or P3 segments of the posterior cerebral artery (PCA) (eFigure 1 in [Supplement 4](#)). Of note, the occlusion site was not adjudicated by a core laboratory to reflect the real-life practice. However, there was a training session during the opening of each center for clarification of the target medium or distal occlusions and especially the distinction between proximal and distal M2 segments.

Main noninclusion criteria were tandem occlusion, secondary medium or distal occlusion (ie, medium or distal occlusion secondary to the migration of a proximal clot), preexisting hemorrhage in the brain tissue fed by the target vessel, stenosis or presence of a stent proximal to the thrombus, and prestroke modified Rankin Scale (mRS) score greater than 1 ([Supplement 5](#)).

Randomization

Patients were randomly assigned in a 1:1 ratio to either the thrombectomy group or the control group using a centralized,

web-based system (CLEANWEB). A minimization algorithm was used to ensure balance between groups according to study center, age (<80 vs ≥80 years), NIHSS score at inclusion (<10 vs ≥10), occlusion site (distal MCA, ACA, and PCA), and IVT initiated or being initiated at randomization vs not planned. To prevent predictability, a probability of 0.9 to assign treatment that minimized imbalance was used and randomization was simple for the first 10 patients before applying the algorithm.

Interventions

In the experimental group, patients underwent thrombectomy, which should start as soon as possible and within 2 hours from randomization (especially for the patients whose care was managed in a drip-and-ship paradigm; ie, admission in a primary stroke center, then transfer to a comprehensive stroke center), under general anesthesia, conscious sedation, or local anesthesia, depending on patient status and local protocols. Thrombectomy was performed using a stent retriever designed for medium or distal vessels, an aspiration catheter, or a combination of both (Supplement 3), depending on technical conditions and at the operator's discretion. Noncontrast brain imaging was performed within 24 hours to rule out a complication.

Patients in both groups received best medical treatment, which could include IVT when not contraindicated and within the appropriate time window (ie, ≤4.5 hours or ≤9 hours in accordance with the EXTEND study criteria²⁵). The choice of thrombolytic agent was left to the treating neurologist. If substantial clinical improvement according to the physician was observed after IVT, thrombectomy could be withheld.

In the control group, patients received best medical treatment alone.

Outcomes

The primary end point was the rate of good clinical outcome, defined as an mRS score of 0 to 2 at 3 months, evaluated by an independent and certified assessor blinded to randomization group.

The secondary end points were (1) the rate of recanalization of the occluded vessel on vascular brain imaging within 48 hours, defined as an arterial occlusion lesion (AOL) score 2 or greater, for both thrombectomy and medical groups, that was evaluated by an independent core laboratory blinded to the randomization group; (2) the rate of angiographic reperfusion in the experimental group at the end of the thrombectomy using the modified Thrombolysis in Cerebral Infarction (TICI) score (successful reperfusion: score 2b-3) and the extended TICI score (successful reperfusion: score ≥2b67) on the control digital subtraction angiography at the end of thrombectomy (the modified and extended TICI scores were adapted to the territory fed by the target vessel occluded); (3) the rate of excellent clinical outcome (defined as mRS score ≤1) at 3 months; (4) all adverse and serious adverse events with a particular attention to perforation rate, embolus migration to a new territory, symptomatic intracranial hemorrhage (including symptomatic subarachnoid hemorrhage) according to the ECASS II (European Co-operative Acute Stroke Study II) crite-

ria, subarachnoid hemorrhage (symptomatic and asymptomatic); and (5) mortality rate at 3 months.

Sample Size Calculation

According to previous studies, patients treated with best medical treatment alone for an AIS related to an MDVO had a rate of good clinical outcome ranging from 20% to 77%,^{26,27} with a weighted average of 40%. In the first 5 RCTs evaluating thrombectomy in proximal occlusions, the minimal difference observed with thrombectomy was 13%.³⁻⁷ To show an absolute difference of 13% between the 2 groups with a global α risk of .05, a power of 0.8, and a bilateral test, we estimated a total number of patients to be randomized of 488 patients, accounting for 5% of follow-up and the conduct of the interim analysis.

Statistical Analysis

Baseline characteristics are described with median (IQR) for quantitative variables and number (percentage) for qualitative variables.

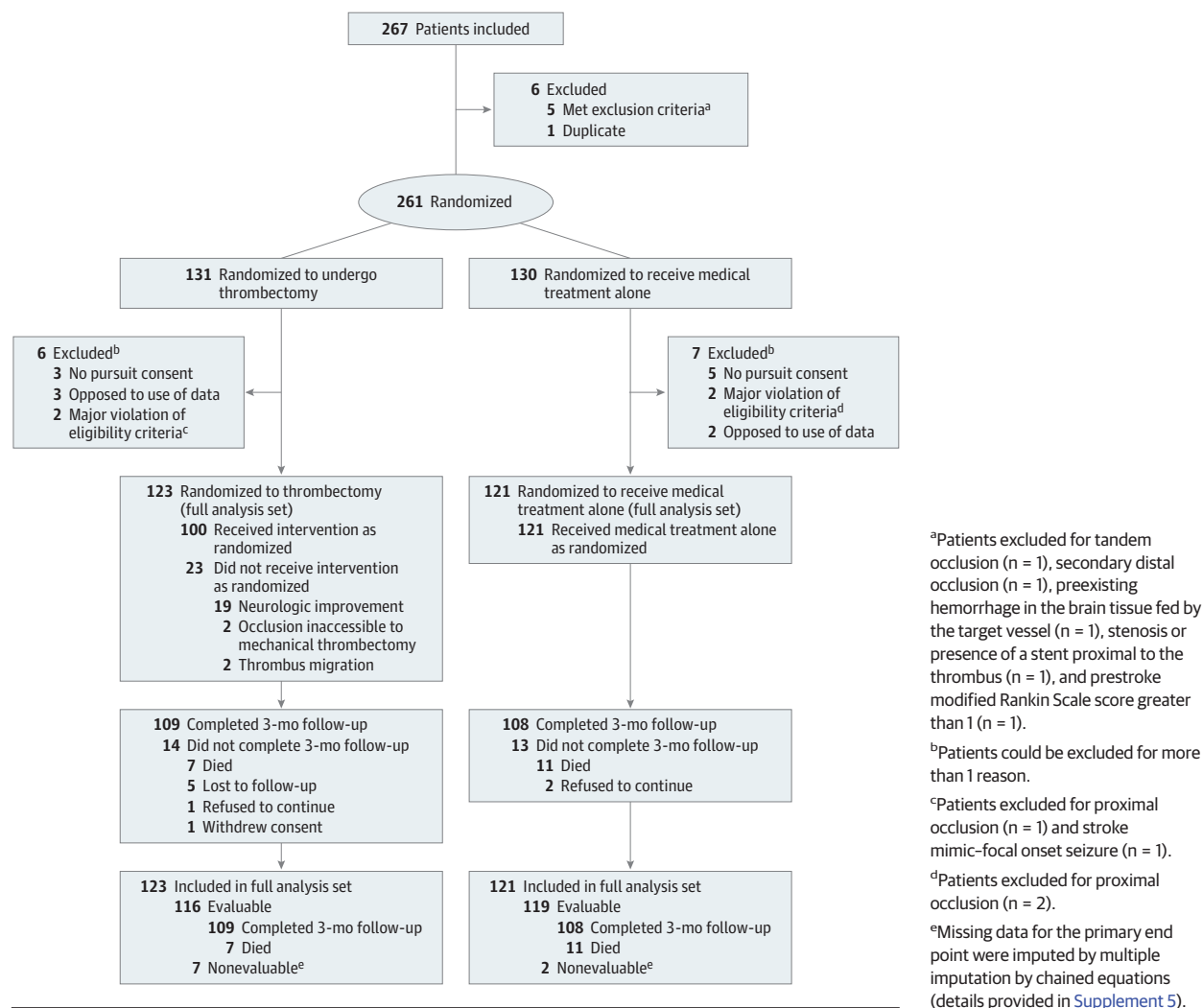
We planned a sequential analysis of the primary end point according to the Lan and DeMets method, with the O'Brien & Fleming spending function with 2 analyses: an interim analysis after the evaluation of the primary end point for the first 163 randomized patients (one-third of the planned sample size) and a final analysis. The trial could be stopped after the interim analysis in case of a difference between the 2 groups attested at the α threshold of .0002 or for futility in case of conditional power less than 30%. After assessment of the results of the interim analysis on December 8, 2024, the data and safety monitoring board recommended to stop the trial for the following reasons: a lower efficacy for the experimental group, although not reaching the predefined α risk threshold; overrepresentation of symptomatic intracranial hemorrhages in the experimental group; and futility with conditional power less than 10%. The trial ended with the follow-up of the last included patient.

The primary analysis was the comparison of the rate of good clinical outcome at 3 months between the 2 groups with a mixed logistic regression model accounting for the minimization factors age, NIHSS score at inclusion, occlusion site, and IVT initiated or planned at inclusion with randomization group as fixed effects and center as a random effect. An adjusted absolute risk difference was also estimated, as was its 95% CI by using bootstrap. The primary analysis was performed in the full analysis set including all patients in the group in which they were randomized, excluding those who opposed to the use of their data and those included in emergency situations without pursuit consent in accordance with French legislation. We also excluded patients who were incorrectly enrolled and had a different condition (eg, stroke due to proximal occlusion). Missing data for the primary end point were imputed by multiple imputation by chained equations (Supplement 5).

The significance threshold of the interim analysis had limited impact on the α risk. Therefore, the significance threshold for the final analysis was .05.

We planned 4 subgroup analyses for the primary end point according to age (<80 vs ≥80 years), NIHSS score at inclusion

Figure 1. Flow Diagram for the DISCOUNT Trial



(<10 vs ≥10), occlusion site on baseline imaging (distal MCA and ACA vs PCA) and IVT initiated or planned at inclusion (yes vs no). Interaction was tested in the mixed logistic regression model.

We conducted the following sensitivity analyses for the primary end point: (1) a per-protocol analysis excluding patients who did not receive the intervention allocated by randomization and those who prematurely withdrew from the study; and (2) the full analysis set additionally excluding patients with a prestroke mRS score of 1 or greater.

Details of the analysis for secondary end points are reported in Supplement 5. All safety analyses were conducted according to the intervention actually received.

We did not correct subgroup or secondary end points analyses for multiplicity of tests. Therefore, these analyses should be considered exploratory.

Statistical analysis was performed with R version 4.4.2 (R Foundation for Statistical Computing).²⁸ All statistical tests were 2-sided; $P < .05$ was considered statistically significant. The full statistical analysis plan is in Supplement 2.

Results

Patients

From November 2021 to November 2024, 261 patients were randomized (Figure 1). Of these, 8 were randomized in an emergency and the patient or his or her legal representative did not sign the pursuit consent and 5 withdrew consent with an opposition to the use of their data. We also excluded 4 patients with a major violation of the trial indication (3 had an AIS due to a proximal occlusion and 1 had a stroke-mimic focal onset seizure) resulting in 244 patients in the full analysis set: 123 randomized to the thrombectomy group and 121 to the control group (eTable 1 in Supplement 4); of these, 217 (89%) completed follow-up. Follow-up ended in April 2025.

The baseline characteristics were mostly balanced between the 2 groups (Table 1). The median age was 75 years (IQR, 67-81) and 56% were male. The median NIHSS score at randomization was 8 (IQR, 6-12), and 39% of patients presented an admission NIHSS score of 10 or greater. The occlusion site

was distal MCA in 75% of patients, followed by PCA in 19% and ACA in 7%. IVT was initiated or planned to be initiated at randomization in 66% of patients.

Intervention

Thrombectomy was performed in 100 of 123 patients (81%) in the intervention group, with a median time since symptom onset of 4.3 hours (IQR, 3.3-5.6) and a median time since randomization of 0.7 hours (IQR, 0.4-1). Nineteen patients did not undergo thrombectomy because of early neurologic improvement (all except 1 underwent IVT), 2 had a thrombus migration, and 2 had an occlusion site that was not accessible to thrombectomy. The median number of passes performed was 1.5 (IQR, 1-2). Details of the procedure are reported in eTable 2 in Supplement 4. At the end of the thrombectomy, 85% of patients (95% CI, 76%-91%) had a reperfusion based on a modified TICI scale of 2b or higher, 79% (95% CI, 69%-86%) had an extended TICI score of 2b67 or greater, and 68% (95% CI, 58%-77%) had an extended TICI score of 2c-3.

Primary End Point

A good clinical outcome at 3 months was observed in 72 of 116 patients (62%) in the thrombectomy group and in 81 of 119 patients (68%) in the control group. After multiple imputation of missing data, the adjusted odds ratio was 0.73 (95% CI, 0.40-1.31; $P = .29$), with an adjusted absolute difference of -6.8% (95% CI, -19.4% to 5.7%) (Table 2). The distribution of the mRS score at 3 months is shown in Figure 2A.

The subgroup analyses did not show significant heterogeneity by age, NIHSS score at baseline, occlusion site, and initiated or planned IVT at inclusion (Figure 2B).

Sensitivity analyses yielded consistent results (eFigure 2 in Supplement 4). In particular, in the per-protocol analysis, the adjusted absolute difference was -10.1% (95% CI, -23.6% to 4.0%).

Secondary End Points

Of the 154 patients with adequate imaging available, 85% of patients in the thrombectomy group and 73% in the control group (difference, 11.5% [95% CI, 0.10%-23.1%]; $P = .049$) had a recanalization based on AOL score evaluated on the 48-hour control imaging by the core laboratory. An excellent clinical outcome was observed in 41% of patients in the thrombectomy group and in 36% of patients in the control group (difference, 5.2% [95% CI, -7.0% to 17.3%]; $P = .41$) (Table 2).

The number of adverse events and serious adverse events were significantly higher in patients who received thrombectomy than in those who did not (Table 3). Symptomatic intracranial hemorrhages were significantly more frequent with thrombectomy (11% vs 3%; difference, 8.2% [95% CI, 4.3%-17.9%]; $P = .008$), as were subarachnoid hemorrhages of any type (symptomatic or asymptomatic) (13% vs 2%; difference, 10.9% [95% CI, 4.3%-17.9%]; $P < .001$) and embolus migration in a new territory (5% vs 1%; difference, 4.3% [95% CI, 0.30%-9.3%]; $P = .04$). The mortality rate did not significantly differ between the 2 groups (6% vs 8%; difference, -2.3% [95% CI, -8.6% to 4.1%]; $P = .49$).

Table 1. Characteristics of the Patients at Baseline

Baseline characteristics	Thrombectomy with medical treatment (n = 123)	Medical treatment alone (n = 121)
Age		
Median (IQR), y	75 (67-81)	75 (66-81)
≥80 y, No. (%)	39 (32)	37 (31)
Sex, No. (%)		
Male	69 (56)	68 (56)
Female	54 (44)	53 (44)
Prestroke modified Rankin Scale score, No. (%)		
0 or 1	120 (98)	118 (98)
>1	3 (2)	3 (2)
Medical history, No. (%)		
Hypertension	87 (71)	62 (51)
Hyperlipidemia	52 (42)	43 (36)
Diabetes	30 (24)	14 (12)
Atrial fibrillation	26 (21)	20 (17)
Ischemic heart disease	23 (19)	24 (20)
Previous stroke	14 (11)	15 (12)
Smoking status, No. (%)		
Current	14 (12)	25 (23)
Past	34 (29)	28 (25)
NIHSS score at presentation		
Median (IQR)	9 (6-12)	8 (6-12)
Score ≥10, No. (%)	47 (38)	47 (39)
Imaging modality for AIS diagnosis, No. (%)		
MRI/MRA	100 (82)	95 (79)
CT-CTA	22 (18)	26 (21)
Site of occlusion on CTA/MRA, No. (%)		
Distal M2 segment of MCA	72 (59)	77 (64)
M3 segment of MCA	18 (15)	15 (12)
A1, A2, A3 segments of ACA	9 (7)	7 (6)
P1, P2, P3 segments of PCA	24 (20)	22 (18)
DWI-ASPECTS score on MRI, median (IQR)	8 (7-9) [n = 90]	8 (7-9) [n = 78]
ASPECTS score on CT, median (IQR)	10 (9-10) [n = 20]	9 (9-10) [n = 23]
Initiated or planned intravenous thrombolysis treatment, No. (%)	81 (66)	81 (67)
Interval between onset and randomization, median (IQR), h	3.5 (2.5-4.7)	3.3 (2.5-4.7)
Interval between onset and intravenous thrombolysis, median (IQR), h ^a	2.7 (2.2-3.5)	2.9 (2.3-3.5)
Interval between onset and intravenous thrombolysis, No. (%) ^a		
Within 4.5 h	67 (92)	74 (93)
4.5-9 h	4 (5)	6 (8)
≥9 h	2 (3)	0
Interval time between onset and arterial puncture, median (IQR), h ^b	4.3 (3.3-5.6)	NA

Abbreviations: ACA, anterior cerebral artery; AIS acute ischemic stroke; ASPECTS, Alberta Stroke Program Early CT Score; CT, computed tomography; CTA, computed tomography angiography; DWI, diffusion-weighted imaging; MCA, middle cerebral artery; MRA, magnetic resonance angiography; MRI, magnetic resonance imaging; NA, not applicable; NIHSS, National Institutes of Health Stroke Scale; PCA, posterior cerebral artery.

^a Thirteen missing data: 8 for the experimental group, 5 for the control group.

^b Two missing data.

Table 2. Efficacy End Points

Outcome	No./total (%)		Adjusted odds ratio (95% CI)	P value	Absolute difference, % (95% CI)
	Thrombectomy with best medical treatment (n = 123)	Medical treatment alone (n = 121)			
Primary end point					
Good clinical outcome (mRS 0-2) at 3 mo	72/116 (62) ^a	81/119 (68) ^a	0.73 (0.40 to 1.31) ^b	.29	-6.8 (-19.4 to 5.7) ^c
Secondary efficacy end points					
Recanalization of occluded vessel ^d	83/98 (85)	71/97 (73)		.05	
Angiographic reperfusion after thrombectomy on DSA ^e					
Extended TICI 2b50-3 ^f	83/98 (85)	NA			
Extended TICI 2b67-3 ^f	77/98 (79)	NA			
Excellent clinical outcome (mRS 0-1) at 3 mo	48/116 (41)	43/119 (36)		.41	

Abbreviations: DSA, digital subtraction angiography; mRS, modified Rankin Scale; NA, not applicable; TICI, Thrombolysis in Cerebral Infarction.

^a The mRS was assessed by a blinded independent assessor. Case numbers and percentages are reported based on observed values before multiple imputation.

^b Adjusted odds ratio estimated using a mixed logistic regression model adjusted on the minimization factors of age (<80 vs ≥80 years), National Institutes of Health Stroke Scale score at inclusion (<10 vs ≥10), occlusion site (distal middle cerebral artery, anterior cerebral artery, or posterior cerebral artery), and intravenous thrombolysis initiated or planned at inclusion, with randomization group as fixed effect and center as a random effect in the full analysis set after multiple imputation.

^c Absolute difference adjusted on minimization factors, namely age (<80 vs ≥80 years), National Institutes of Health Stroke Scale score at inclusion (<10 vs ≥10), occlusion site (distal middle cerebral artery, anterior cerebral artery, or posterior cerebral artery), and intravenous thrombolysis initiated or planned at inclusion, with randomization group as fixed effect, and center as a

random effect in the full analysis set after multiple imputation. Confidence interval was estimated using bootstraps.

^d The recanalization of the occluded vessel is defined as an AOL (arterial occlusive lesion) score 2 or greater on imagery with injection within 24 to 48 hours as assessed by an independent and blinded core laboratory. There was no adequate imaging available for 25 patients randomized to the experimental group and 24 patients randomized to the control group.

^e Of the 123 patients randomized to the thrombectomy group, thrombectomy was performed in only 100 and thrombectomy was stopped in 2 additional patients, resulting in 98 patients for whom angiographic reperfusion on DSA could be assessed.

^f Extended TICI score 0 indicates no recanalization; 1, minor recanalization with no reperfusion; 2a, reperfusion less than 50% of the territory supplied by the occluded vessel; 2b50, reperfusion of 50% to 66% of the target territory; 2b67, reperfusion of 67% to 89% of the target territory; 2c, reperfusion of 90% to 99% of the target territory; 3, full reperfusion (100%).

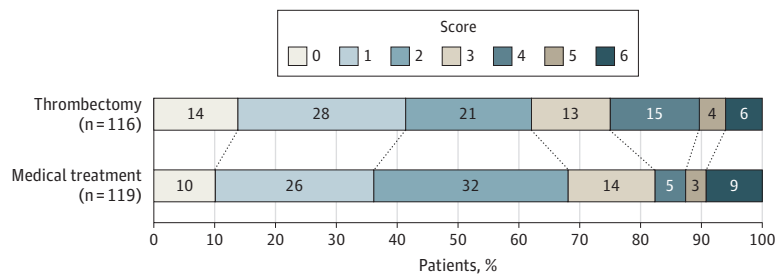
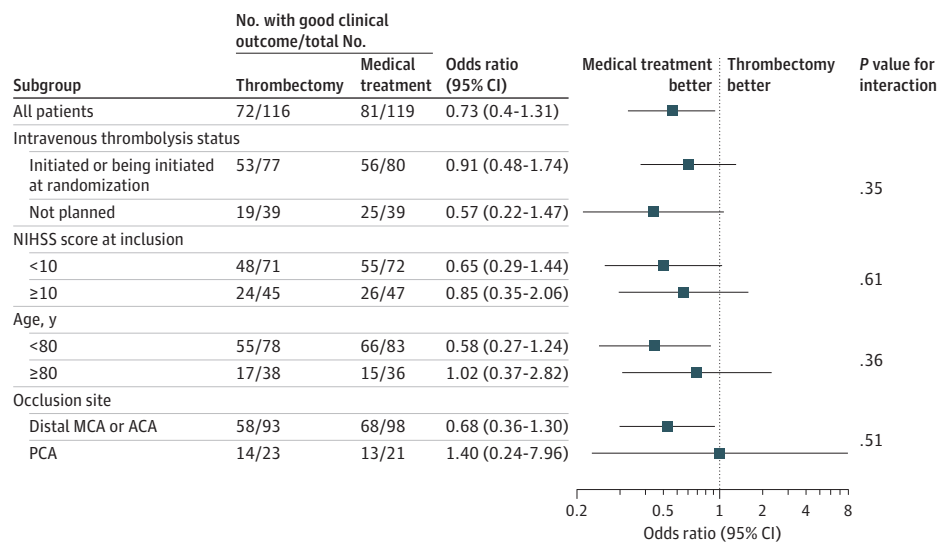
Discussion

In this multicenter RCT, thrombectomy did not improve the rate of good clinical outcome at 3 months among patients with an AIS due to a medium or distal vessel occlusion. Although the rate of recanalization seemed higher with thrombectomy than with the best medical treatment alone, thrombectomy was associated with a greater number of adverse and serious adverse events. Symptomatic intracranial hemorrhage, subarachnoid hemorrhage, and embolus migration occurred significantly more often in patients who received thrombectomy. Taken together, these findings have important implications for clinical practice, indicating an increased hemorrhagic risk without functional benefit of thrombectomy in medium or distal vessel occlusions, and therefore not supporting its routine use in this setting.

Unlike the 2 recently published RCTs on medium or distal occlusions thrombectomy,^{19,20} in which the nondominant M2 segment in DISTAL and the proximal M2 segment in ESCAPE-MeVO were included, this study did not consider proximal M2 occlusions. In both DISTAL¹⁹ and ESCAPE-MeVO,²⁰ the primary end point was the distribution of the mRS score at 3 months. Excellent clinical outcome (mRS score 0-1) and good clinical outcome (mRS score 0-2) were defined as secondary end points in these studies. Both trials reported no significant difference between thrombectomy and best medical treatment alone for these functional outcomes. Results in the current trial

were consistent, with a slightly higher proportion of patients achieving a good clinical outcome in the control group (68%) compared with DISTAL (55%) and ESCAPE-MeVO (59%). This may be explained by the choice to exclude proximal M2 occlusions, which may be associated with a poorer prognosis than more distal M2-M3 occlusions.

There are possible explanations regarding the lack of significant benefit for thrombectomy in MDVOs. First, the findings suggest that best medical treatment alone including IVT was already highly effective in patients with MDVOs. Although the recanalization rate was significantly higher with thrombectomy in the current trial, this did not translate into a better clinical outcome at 3 months, probably due to the higher rates of symptomatic hemorrhages. Indeed, symptomatic intracranial hemorrhages occurred more than 3 times more frequently with thrombectomy, reflecting a higher procedural risk for medium or distal occlusions. This may be related to catheter or microguidewire navigation or to traction on medium or distal vessels during clot retrieval, potentially resulting in avulsion of perforating branches. In the DISTAL¹⁹ and ESCAPE-MeVO²⁰ trials, the rates of symptomatic intracranial hemorrhage were more than 2 times higher in the thrombectomy group than in the control group (5.9% vs 2.6% and 5.4 vs 2.2%, respectively). These results highlight the technical difficulties of conducting thrombectomy in medium or distal vessel occlusions. To improve the effectiveness of medium or distal thrombectomy, it may be necessary to optimize the devices currently

Figure 2. Distribution Diagram of Modified Rankin Scale Score at 3 Months and Dot-and-Whisker Plot of Subgroup Analyses for Rate of Good Clinical Outcome at 3 Months**A** Distribution of Modified Rankin Scale score at 3 mo**B** Subgroup analyses for the rate of good clinical outcome (score 0-2) at 3 mo

available, which are, as of now, merely smaller versions of those used for LVOs.

Second, patients in the DISTAL,¹⁹ ESCAPE-MeVO,²⁰ and DISCOUNT trials were relatively older, with a median age around 75 years. Median NIHSS scores at randomization were low (8, 6, and 8, respectively), which may not reflect the broader population of patients with an AIS. Younger patients with more severe presentation may have been under-represented because some investigators may have considered randomization to the control group an unethical loss of opportunity.

Third, the median time from symptom onset to thrombectomy initiation was long across all 3 trials: 294 minutes for DISTAL,¹⁹ 311 minutes for ESCAPE-MeVO,²⁰ and 258 minutes for DISCOUNT. Longer delays may reduce the benefit of thrombectomy, because for medium or distal vessel occlusions, the chance to maintain a sufficient perfusion level via leptomeningeal collaterals is lower than in proximal occlusions.²⁹

Last, although mRS is the standard end point in AIS trials, it may not fully capture subtle clinical improvement for medium or distal occlusions.³⁰

None of the subgroup analyses (based on stroke severity or IVT) in the 3 RCTs suggested that a particular subgroup

Table 3. Safety Outcomes According to the Treatment Received

Outcome	Thrombectomy received (n = 100)	No thrombectomy receive (n = 144)
Serious adverse events, No. (%)	36 (36)	38 (26)
No. of serious adverse events, median (IQR)	0 (0-1)	0 (0-1)
Mortality at 3 mo, No. (%)	6 (6)	12 (8)
Adverse events, No. (%)	95 (95)	122 (85)
Number of adverse events, median (IQR)	4 (2-8)	3 (1-5)
Adverse events of interest, No. (%)		
Vessel perforation rate	3 (3)	0
Embolus migration in a new territory	5 (5)	1 (1)
Symptomatic intracranial hemorrhage within 7 d	11 (11)	4 (3)
Subarachnoid hemorrhage ^a	13 (13)	3 (2)

^a Symptomatic or asymptomatic.

may benefit from thrombectomy. Recently, 2 trials, not yet published, were presented at the 2026 International Stroke Conference: ORIENTAL-MeVO²¹ (NCT06146790) and DISTALS²² (NCT05152524). Both trials showed positive results

for thrombectomy. However, in these RCTs, the rate of IVT in the medical treatment group was low (ORIENTAL-MeVO) or absent (DISTALS). Other RCTs, such as FRONTIER-AP (ACTRN12621001746820), are ongoing. An individual patient data meta-analysis including all randomized trials, with subgroup analyses by site of vessel occlusion and other relevant variables, would be valuable to identify patient subgroups who might benefit from thrombectomy and those for whom it should be avoided.

Limitations

First, the median NIHSS score of 8 and the high IVT rate suggest a study population with moderate symptoms severity, which may limit the external validity of the findings for patients with more severe MDVOs or those ineligible for IVT. It is possible that some patients with more severe occlusions, who are perceived by treating physicians as more likely to benefit, were treated with thrombectomy outside the trial.³¹ Second, a potential lack of statistical power due to the premature termination of the trial and the higher-than-expected rate of good clinical outcome in the control group cannot be ruled out. In addition, the assumed 13% absolute difference planned in sample size calculation may be seen as optimistic

for medium or distal vessel occlusions. Third, due to French legal requirements, only data from patients who provided consent could be analyzed, precluding an intention-to-treat analysis in the emergency context. Nevertheless, despite these exclusions, patient characteristics were relatively well-balanced in the 2 groups. Fourth, the evaluation of the reperfusion status at the end of the thrombectomy procedure was site-reported, which can be a source of bias leading to an overestimation of the reperfusion rate. Fifth, not all patients underwent follow-up imaging at 48 hours, resulting in missing AOL scores. This was partly due to the drip-and-ship care management of some patients, which made it difficult to verify completion of follow-up imaging after transfer back to the primary stroke center.

Conclusions

In this multicenter RCT involving patients with AIS due to a medium or distal vessel occlusion, thrombectomy did not increase the rate of good clinical outcome at 3 months. There was an increased rate of hemorrhagic complications associated with thrombectomy.

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Author Affiliations: AP-HP, Sorbonne Université, Hôpital Pitié-Salpêtrière, Department of Neuroradiology, Pitié-Salpêtrière Hospital, Paris, France (Clarençon); GRC NoVA, Sorbonne Université, Paris, France (Clarençon); AP-HP, Department of Neuroradiology, Tours University Hospital, Tours, France (Bala); Sorbonne Université, Hôpital Pitié-Salpêtrière, Unité de recherche clinique, Paris, France (Marnat); Department of Vascular Neurology, Tours University Hospital, Tours, France (Pasi); Department of Neuroradiology, Bordeaux University Hospital, Bordeaux, France (Marnat); Department of Vascular Neurology, Bordeaux University Hospital, Bordeaux, France (Sibon); Department of Diagnostic and Therapeutic Neuroradiology, INSERM U1254, CHRU-Nancy, Université de Lorraine, Nancy, France (Gory); Department of Neurology, INSERM U1116, Stroke Unit, CHRU-Nancy, Université de Lorraine, Nancy, France (Richard); Department of Neuroradiology, Strasbourg University Hospital, Strasbourg, France (Pop); Department of Vascular Neurology, Strasbourg University Hospital, UR3072, CRBS, University of Strasbourg, Strasbourg, France (Wolff); Department of Neuroradiology, Poitiers University Hospital, Poitiers, France (Fauché); Department of Vascular Neurology, Poitiers University Hospital, Poitiers, France (Lamy); Department of Neuroradiology, Montpellier University Hospital, Montpellier, France (Costalat); Department of Vascular Neurology, Montpellier University Hospital, Montpellier, France (Arquizaan); Institute of Psychiatry and Neuroscience of Paris (IPNP), INSERM U1266, Paris, France (Arquizaan, Rodriguez-Régent, Turc, Alamowitch); Department

of Neuroradiology, GHU Paris Psychiatrie et Neurosciences, Paris, France (Rodriguez-Régent); Department of Vascular Neurology, GHU Paris Psychiatrie et Neurosciences, Paris, France (Turc); Université Paris Cité, Paris, France (Turc); Department of Neuroradiology, Toulouse University Hospital, Toulouse, France (Cognard); Department of Vascular Neurology, Toulouse University Hospital, Toulouse, France (Fontaine); Department of Neuroradiology, Foch Hospital, Suresnes, France (Di Maria); Department of Vascular Neurology, Foch Hospital, Suresnes, France (Lapergue); Department of Neuroradiology, Lille University Hospital, Lille, France (Bricout); Department of Vascular Neurology, Lille University Hospital, Lille, France (Henon); Department of Neuroradiology, Henri-Mondor Hospital, Créteil, France (Tuilier); Department of Vascular Neurology, Henri-Mondor Hospital, Créteil, France (Wittwer); Department of Neuroradiology, Nantes University Hospital, Nantes, France (Alexandre); Department of Vascular Neurology, Nantes University Hospital, Nantes, France (De Gaalon); Department of Interventional Neuroradiology, Kremlin-Bicêtre Hospital, Kremlin-Bicêtre, France (Spelle); Department of Vascular Neurology, Kremlin-Bicêtre Hospital, Kremlin-Bicêtre, France (Denier); Department of Neuroradiology, Clermont-Ferrand University Hospital, Clermont-Ferrand, France (Chabert); Department of Vascular Neurology, Clermont-Ferrand University Hospital, Clermont-Ferrand, France (Ferrier); Department of Interventional Neuroradiology, Limoges University Hospital, Limoges, France (Rouchaud); Department of Neurology, Limoges University Hospital, Limoges, France (Macian-Montoro); Department of Neuroradiology, Reims University Hospital, Reims, France (Soize); Department of Vascular Neurology, Reims University Hospital, Reims, France (Moulin); Department of Neuroradiology, Fondation A de Rothschild, Paris, France (Boisseau); Department of Vascular Neurology, Fondation A de Rothschild,

Paris, France (Sabben); Department of Neuroradiology, Caen University Hospital, Caen, France (Barbier); Department of Vascular Neurology, Caen University Hospital, Caen, France (Boulanger); Department of Neuroradiology, Lyon University Hospital, Lyon, France (Eker); Department of Vascular Neurology, Lyon University Hospital, Lyon, France (Cho); Department of Neuroradiology, Brest University Hospital, Brest, France (Gentric); Department of Vascular Neurology, Brest University Hospital, Brest, France (Timsit); Department of Neuroradiology, Lariboisière Hospital, Paris, France (Labeyrie); Department of Vascular Neurology, Lariboisière Hospital, Paris, France (Reiner); Department of Vascular Neurology, Versailles Hospital, Versailles, France (Pico); Université de Paris, CRESS, INSERM, INRA, URCEco, AP-HP, Hôpital de l'Hôtel Dieu (Durand-Zaleski); Department of Vascular Neurology, Pitié-Salpêtrière Hospital, Paris, France (Alamowitch); Sorbonne Université, INSERM, Institut Pierre Louis d'Epidémiologie et de Santé Publique, AP-HP, Hôpital Pitié-Salpêtrière, Département de Santé Publique, Paris, France (Dechartres).

Author Contributions: Drs Clarençon and Dechartres had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. *Concept and design:* Clarençon, Lamy, Costalat, Durand-Zaleski, Dechartres.

Acquisition, analysis, or interpretation of data: Clarençon, Bala, Baptiste, Pasi, Marnat, Sibon, Gory, Richard, Pop, Wolff, Fauché, Costalat, Arquizaan, Rodriguez-Régent, Turc, Cognard, Fontaine, Di Maria, Lapergue, Bricout, Henon, Tuilier, Wittwer, Alexandre, De Gaalon, SPELLE, Denier, Chabert, Ferrier, Rouchaud, Macian-Montoro, Soize, Moulin, Boisseau, Sabben, Barbier, Boulanger, Eker, Cho, Gentric, Timsit, Labeyrie, Reiner, Pico, Alamowitch, Dechartres.

Drafting of the manuscript: Clarençon, Baptiste, Lamy, Costalat, Rodriguez-Régent, Alexandre, Barbier, Eker, Durand-Zaleski, Dechartres. **Critical review of the manuscript for important intellectual content:** Bala, Pasi, Marnat, Sibon, Gory, Richard, Pop, Wolff, Fauché, Costalat, Arquizan, Turc, Cognard, Fontaine, Di Maria, Lapergue, Bricout, Henon, Tuilier, Wittwer, De Gaalon, SPELLE, Denier, Chabert, Ferrier, Rouchaud, Macian-Montoro, Soize, Moulin, Boisseau, Sabben, Boulanger, Eker, Cho, Gentric, Timsit, Labeyrie, Reiner, Pico, Alamowitch. **Statistical analysis:** Baptiste, Durand-Zaleski, Dechartres. **Obtained funding:** Clarençon, Costalat, Dechartres. **Administrative, technical, or material support:** Bala, Marnat, Fauché, Lamy, Costalat, Lapergue, Wittwer, Alexandre, SPELLE, Denier, Chabert, Macian-Montoro, Boulanger, Gentric, Pico. **Supervision:** Clarençon, Bala, Richard, Pop, Wolff, Costalat, Arquizan, Cognard, Barbier, Alamowitch, Dechartres.

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Group Information: A list of the DISCOUNT Investigators appears in [Supplement 6](#). Steering committee: *Members of the committee:* Frédéric Clarençon MD, PhD (investigator), Agnes Dechartres MD, PhD (methodologist), representatives of the sponsor (DRCI and URC). **Missions:** Define the overall structure of the study, coordinate information, determine the initial methodology and oversee the trial. Propose procedures to be followed during the study. Scientific committee: *Members of the committee:* Frédéric Clarençon MD, PhD, Agnès Dechartres MD, PhD, Sonia Alamowitch MD, Raphael Blanc MD, Sandrine Deltour MD. **Missions:** determine the objective, write the protocol, recommend changes to the protocol during the trial and write the article.

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